

IoT Automated Smart Irrigation and Soil Moisture Analytics System

Project Synopsis & Executive Proposal (MCA Mini Project)

Academic Year:	2024-2025 (Semester 3)
Course / Type:	Mini Project (MCA-305)
Department:	Master of Computer Applications (MCA)
Repository URL:	https://github.com/projectvault/smart-irrigation-iot.git
Hardware Target:	ESP32-WROOM-32, Capacitive Soil Moisture Sensors, 12V Solenoid Valves

1. Project Summary & Motivation

Agricultural irrigation consumes over 70% of global accessible freshwater. Traditional timer-based irrigation schedules often result in substantial water wastage due to disregard for actual soil water retention and real-time precipitation dynamics.

This project develops an IoT Automated Smart Irrigation and Soil Moisture Analytics System deploying ESP32 microcontrollers and capacitive moisture sensors. By collecting localized soil telemetry and computing evapotranspiration thresholds, the system activates solenoid valve actuators over MQTT only when moisture falls below critical crop thresholds.

2. Core Objectives

- Deploy low-power ESP32 edge microcontroller nodes across segmented field zones.
- Continuously measure ground moisture percentage, ambient humidity, and thermal levels.
- Automate solenoid valve relays with safety timeout shut-offs to prevent over-watering.
- Stream real-time telemetry to a cloud dashboard for agricultural analytics and water volume savings tracking.

3. System Architecture & Methodology

The edge nodes acquire analog telemetry via capacitive moisture sensors (v1.2) and DHT22 sensors. An embedded FreeRTOS loop converts raw voltage signals into calibrated volumetric water content (VWC) metrics.

Data packets are packaged in JSON and transmitted over 2.4 GHz Wi-Fi / MQTT broker to the ProjectVault analytics backend. When threshold $VWC < 35\%$ is met, a relay activation command is triggered for 120 seconds.

4. Expected Outcomes & Impact

- Conservation of irrigation water by approximately 30% to 38% compared to timer systems.
- Reduction in crop root rot caused by oversaturation in clay and loam soils.
- Automated alert dispatch via Telegram/Email when anomalous sensor disconnects occur.